

Homework 1

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Exercise 1

Part (a)

$$\text{col} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix is already in row echelon form. Since there are 2 pivot columns, the column space has dimension 2.

Part (b)

$$\text{span}\{(1, 1, 1), (1, -1, 1), (-1, 1, 1), (1, 1, -1)\}$$

The dimension of the span is equal to the dimension of:

$$\text{col} \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

Putting the matrix in row echelon form using Gaussian elimination:

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix} \begin{array}{l} \leftarrow -1 \\ \leftarrow + \\ \leftarrow + \end{array} \begin{array}{l} -1 \\ -1 \\ + \end{array}$$
$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & 2 & -2 \end{bmatrix}$$

Since there are 3 pivot columns, the column space and the span have dimension 3.

Part (c)

$$\text{span}\{(2, 7, 9), (3, 5, 1), (0, 1, 0)\}$$

The dimension of the span is equal to the dimension of:

$$\text{col} \begin{bmatrix} 2 & 3 & 0 \\ 7 & 5 & 1 \\ 9 & 1 & 0 \end{bmatrix}$$

Putting the matrix in row echelon form using Gaussian elimination:

$$\begin{bmatrix} 2 & 3 & 0 \\ 7 & 5 & 1 \\ 9 & 1 & 0 \end{bmatrix} \begin{array}{l} \left[\begin{array}{c} \leftarrow -\frac{7}{2} \\ \leftarrow + \end{array} \right] -\frac{9}{2} \\ \left[\begin{array}{c} \leftarrow + \end{array} \right] + \end{array}$$
$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & -\frac{11}{2} & 1 \\ 0 & -\frac{19}{2} & 0 \end{bmatrix} \begin{array}{l} \left[\begin{array}{c} \leftarrow \frac{19}{11} \\ \leftarrow + \end{array} \right] \end{array}$$
$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & -\frac{11}{2} & 1 \\ 0 & 0 & \frac{19}{11} \end{bmatrix}$$

Since there are 3 pivot columns, the column space and the span have dimension 3.

Part (d)

$$\text{col} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Putting the matrix in row echelon form using Gaussian elimination:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \left[\begin{array}{c} \leftarrow -1 \\ \leftarrow + \end{array} \right] \end{array}$$
$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \left[\begin{array}{c} \leftarrow \end{array} \right] \\ \left[\begin{array}{c} \leftarrow \end{array} \right] \end{array}$$
$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Since there are 2 pivot columns, the column space has dimension 2.

Exercise 2

Part (a)

$$f(x, y, z) = 0$$

A linear function must preserve vector addition.

$$\begin{aligned}f(x_1, y_1, z_1) + f(x_2, y_2, z_2) &= 0 + 0 = 0 \\f(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= 0\end{aligned}$$

A linear function must preserve scalar multiplication. For $a \in \mathbb{R}$:

$$\begin{aligned}a \cdot f(x, y, z) &= a \cdot 0 = 0 \\f(ax, ay, az) &= 0\end{aligned}$$

Since both vector addition and scalar multiplication are preserved, the function is linear.

Part (b)

$$f(x, y, z) = 1$$

A linear function must preserve vector addition.

$$\begin{aligned}f(x_1, y_1, z_1) + f(x_2, y_2, z_2) &= 1 + 1 = 2 \\f(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= 1\end{aligned}$$

A linear function must preserve scalar multiplication. For $a \in \mathbb{R}$:

$$\begin{aligned}a \cdot f(x, y, z) &= a \cdot 1 = a \\f(ax, ay, az) &= 1\end{aligned}$$

Since neither vector addition nor scalar multiplication are preserved, the function is not linear.

Part (c)

$$f(x, y, z) = (1 + x, 2z)$$

A linear function must preserve vector addition.

$$\begin{aligned}f(x_1, y_1, z_1) + f(x_2, y_2, z_2) &= (1 + x_1, 2z_1) + (1 + x_2, 2z_2) \\&= (2 + x_1 + x_2, 2z_1 + 2z_2) \\f(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= (1 + (x_1 + x_2), 2(z_1 + z_2)) \\&= (1 + x_1 + x_2, 2z_1 + 2z_2)\end{aligned}$$

A linear function must preserve scalar multiplication. For $a \in \mathbb{R}$:

$$\begin{aligned} a \cdot f(x, y, z) &= a \cdot (1 + x, 2z) = (a(1 + x), 2az) \\ &= (a + ax, 2az) \\ f(ax, ay, az) &= (1 + ax, 2az) \end{aligned}$$

Since neither vector addition nor scalar multiplication are preserved, the function is not linear.

Part (d)

$$f(x, y, z) = (x, 2x)$$

A linear function must preserve vector addition.

$$\begin{aligned} f(x_1, y_1, z_1) + f(x_2, y_2, z_2) &= (x_1, 2x_1) + (x_2, 2x_2) \\ &= (x_1 + x_2, 2x_1 + 2x_2) \\ f(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= (x_1 + x_2, 2(x_1 + x_2)) \\ &= (x_1 + x_2, 2x_1 + 2x_2) \end{aligned}$$

A linear function must preserve scalar multiplication. For $a \in \mathbb{R}$:

$$\begin{aligned} a \cdot f(x, y, z) &= a \cdot (x, 2x) = (ax, 2ax) \\ f(ax, ay, az) &= (ax, 2ax) \end{aligned}$$

Since both vector addition and scalar multiplication are preserved, the function is linear.

Part (e)

$$f(x, y, z) = (2x + 3y, x, 0)$$

A linear function must preserve vector addition.

$$\begin{aligned} f(x_1, y_1, z_1) + f(x_2, y_2, z_2) &= (2x_1 + 3y_1, x_1, 0) + (2x_2 + 3y_2, x_2, 0) \\ &= (2x_1 + 2x_2 + 3y_1 + 3y_2, x_1 + x_2, 0) \\ f(x_1 + x_2, y_1 + y_2, z_1 + z_2) &= (2(x_1 + x_2) + 3(y_1 + y_2), x_1 + x_2, 0) \\ &= (2x_1 + 2x_2 + 3y_1 + 3y_2, x_1 + x_2, 0) \end{aligned}$$

A linear function must preserve scalar multiplication. For $a \in \mathbb{R}$:

$$\begin{aligned} a \cdot f(x, y, z) &= a \cdot (2x + 3y, x, 0) = (a(2x + 3y), ax, 0) \\ &= (2ax + 3ay, ax, 0) \\ f(ax, ay, az) &= (2ax + 3ay, ax, 0) \end{aligned}$$

Since both vector addition and scalar multiplication are preserved, the function is linear.

Exercise 3

Let $\mathcal{U}_1, \mathcal{U}_2$ be subspaces of vector space $\mathcal{V}$. Let $u_1, u_2 \in \mathcal{U}_1 \cap \mathcal{U}_2$. Since $u_1, u_2 \in \mathcal{U}_1$, $u_1 + u_2 \in \mathcal{U}_1$. Similarly, $u_1 + u_2 \in \mathcal{U}_2$. Consequently, $u_1 + u_2 \in \mathcal{U}_1 \cap \mathcal{U}_2$, and vector addition is preserved. Let $a \in \mathbb{R}$. Since $u_1 \in \mathcal{U}_1$, $a \cdot u_1 \in \mathcal{U}_1$. Similarly, $a \cdot u_1 \in \mathcal{U}_2$. Consequently, $a \cdot u_1 \in \mathcal{U}_1 \cap \mathcal{U}_2$, and scalar multiplication is preserved. Since $\mathcal{U}_1 \cap \mathcal{U}_2$ is closed under both vector addition and scalar multiplication, $\mathcal{U}_1 \cap \mathcal{U}_2$ is a subspace of $\mathcal{V}$.

However, $\mathcal{U}_1 \cup \mathcal{U}_2$ is not necessarily a subspace of $\mathcal{V}$. Let $\mathcal{U}_1 = \text{span}\{(1, 0)\}$ and $\mathcal{U}_2 = \text{span}\{(0, 1)\}$, which are subspaces of $\mathcal{V} = \mathbb{R}^2$. Consider $u_1 = (1, 0) \in \mathcal{U}_1$ and $u_2 = (0, 1) \in \mathcal{U}_2$. $u_1 + u_2 = (1, 1)$, but since $u_1 + u_2 \notin \mathcal{U}_1$ and $u_1 + u_2 \notin \mathcal{U}_2$, $u_1 + u_2 \notin \mathcal{U}_1 \cup \mathcal{U}_2$. Since $\mathcal{U}_1 \cup \mathcal{U}_2$ is not closed under vector addition, $\mathcal{U}_1 \cup \mathcal{U}_2$ is not a subspace of $\mathcal{V}$.

Exercise 4

$$A = \begin{bmatrix} r_1^T \\ r_2^T \\ \vdots \\ r_m^T \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix}$$

The elements of $A^T A$ can be expressed as:

$$\begin{aligned} (A^T A)_{ij} &= (A^T)_i \cdot (A)_j \\ &= c_i \cdot c_j \end{aligned}$$

The elements of AA^T can be expressed as:

$$\begin{aligned} (AA^T)_{ij} &= (A)_i \cdot (A^T)_j \\ &= r_i \cdot r_j \end{aligned}$$

Exercise 5

Let $A, B \in \mathbb{R}^{m \times n}$ be invertible. Then:

$$\begin{aligned} A^T(A^{-1})^T &= (A^{-1}A)^T \\ &= I^T = I \\ &= A^T(A^T)^{-1} \\ (A^{-1})^T &= (A^T)^{-1} \end{aligned}$$

Further:

$$\begin{aligned} (AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AIA^{-1} = AA^{-1} \\ &= I \\ &= (AB)(AB)^{-1} \\ (B^{-1}A^{-1}) &= (AB)^{-1} \end{aligned}$$

Exercise 6

Part (a)

Let $B \in \mathbb{R}^{m \times k}$ such that $B^T B$ is invertible. Then:

$$\begin{aligned}(B(B^T B)^{-1} B^T)^2 &= (B(B^T B)^{-1} B^T)(B(B^T B)^{-1} B^T) \\ &= B(B^T B)^{-1} (B^T B) (B^T B)^{-1} B^T \\ &= B(B^T B)^{-1} I B^T = B(B^T B)^{-1} B^T\end{aligned}$$

Since $(B(B^T B)^{-1} B^T)^2 = B(B^T B)^{-1} B^T$, $B(B^T B)^{-1} B^T$ is idempotent.

Part (b)

Let A be idempotent. Then:

$$\begin{aligned}(I - A)^2 &= I^2 - IA - AI + A^2 \\ &= I - 2A + A = I - A\end{aligned}$$

Since $(I - A)^2 = I - A$, $I - A$ is idempotent.

Part (c)

Let A be idempotent. Consider $\frac{1}{2}I - A$ and $2I - 4A$. Then:

$$\begin{aligned}\left(\frac{1}{2}I - A\right)(2I - 4A) &= I^2 - 2AI - 2IA + 4A^2 \\ &= I - 4A + 4A = I\end{aligned}$$

Since $(\frac{1}{2}I - A)(2I - 4A) = I$, $\frac{1}{2}I - A$ is invertible, with an inverse of $2I - 4A$.

Part (d)

Let A be idempotent. Let $x \neq 0$ and $\lambda \in \mathbb{R}$ such that $Ax = \lambda x$. Then:

$$\begin{aligned}Ax &= \lambda x \\ A^2 x &= Ax = \lambda Ax \\ \lambda x &= \lambda(\lambda x) = \lambda^2 x \\ (\lambda^2 - \lambda)x &= \lambda(\lambda - 1)x = 0 \\ \lambda &= 0, 1\end{aligned}$$

Exercise 7

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Part (a)

$$\sin(x) \approx x$$

The relative forward error at x is $\left| \frac{\sin(x) - x}{\sin(x)} \right|$. The relative backward error is $\left| \frac{\frac{x^3}{6}}{x} \right|$.

x	rel. forward err.	rel. backward err.
0.1	0.00167	0.00167
0.5	0.04291	0.04167
1.0	0.18840	0.16667

Part (b)

$$\sin(x) \approx x - \frac{x^3}{6}$$

The relative forward error at x is $\left| \frac{\sin(x) - (x - \frac{x^3}{6})}{\sin(x)} \right|$. The relative backward error is $\left| \frac{\frac{x^3}{120}}{x} \right|$.

x	rel. forward err.	rel. backward err.
0.1	8.3452×10^{-7}	8.3333×10^{-7}
0.5	0.00054	0.00052
1.0	0.00967	0.00833

Exercise 8

$$f(x) = \log x$$

Part (a)

The condition number $\kappa_f(x)$ is:

$$\begin{aligned}\kappa_f(x) &= \left| \frac{x \cdot f'(x)}{f(x)} \right| \\ &= \left| \frac{x \cdot \frac{1}{x}}{\log x} \right| = \left| \frac{1}{\log x} \right|\end{aligned}$$

Part (b)

The relative forward error at x is $\left| \frac{f(x+\Delta x) - f(x)}{f(x)} \right|$. The relative backward error is $\left| \frac{\Delta x}{x} \right|$. Using $\Delta x = 10^{-4}$:

x	$f(x)$	$f(x + \Delta x)$	$\kappa_f(x)$
1.005	0.00498	0.00508	200.49958
1.004	0.00399	0.00409	250.49967
1.003	0.00300	0.00310	333.83308
1.002	0.00200	0.00210	500.49983
1.001	0.00100	0.00110	1000.49992

x	rel. forward err.	rel. backward err.	ratio
1.005	0.01994	0.00010	200.48961
1.004	0.02495	0.00010	250.48719
1.003	0.03328	0.00010	333.81644
1.002	0.049948	0.00010	500.47486
1.001	0.09995	0.00010	1000.44995

The relative backward error stays fairly constant (close to 10^{-4}). However, since $|\log x|$ grows dramatically as x approaches 1, the condition number grows dramatically as well, resulting in a large upper bound on the relative forward error and a much larger relative change in $f(x + \Delta x)$ in comparison to $f(x)$.